# CRM Leads Management Product Requirements Document

*Spreadsheet-to-application productization for an embedded CRM module*

Version 1.0 | Based on analysis of the attached workbook and requested future-state workflow

| Snapshot | Decision |
|---|---|
| **Business objective** | Replace spreadsheet lead tracking with a structured CRM module, live calculations, and dashboards. |
| **Primary users** | Sales personnel, sales managers, and an admin user for reference-data maintenance. |
| **Host model** | Single ASP.NET Core web application, hosted by the existing CRM, using JWT authentication. |
| **Authentication note** | The host CRM sends the JWT to the client by postMessage; the client then calls the server with the token as a Bearer token. |
| **Key future-state design** | Customer comes from CRM master data; project is reusable or creatable but unique per customer; formulas become read-only calculated fields. |

Recommended product stance: preserve the business inputs from the workbook, but do not preserve spreadsheet behaviors such as hidden formula columns, grouped header rows, or manual summary rows.

## 1. Executive Summary

The current workbook is serving three jobs at once: lead entry, probability calculation, and management reporting. That is why it has input rows, fixed formula columns, hidden validation cells, and summary rows mixed into the same sheets.

The future-state product should separate those concerns. Sales users should create and maintain structured lead records in a guided form; the system should calculate totals and probabilities on the server; managers should consume dashboards instead of summary rows.

This PRD keeps the business semantics visible in the workbook: existing CRM customers, project names unique per customer, yes/no chance-to-win questions, stage tracking, amount breakdown by work type, due date, perpetual-contract handling, and final outcome status.

## 2. Product Goals and Non-Goals

| Type | Statement |
|---|---|
| **Goal** | Make lead entry fast and consistent inside the CRM. |
| **Goal** | Replace spreadsheet formulas with trusted, server-side calculations. |
| **Goal** | Support both one-time auction opportunities and perpetual / ongoing contracts. |
| **Goal** | Provide live dashboards by salesperson, customer, project, work type, stage, and offer outcome. |
| **Goal** | Support monthly and quarterly forecasting by due date. |
| **Non-goal** | Customer master-data creation or maintenance inside this module. |
| **Non-goal** | Full contract execution, invoicing, or ERP billing lifecycle. |
| **Non-goal** | A standalone login flow outside the host CRM. |

## 3. Current-State Interpretation of the Workbook

The workbook structure is clear enough to derive a strong product model, but it should not be copied literally into the application UI.

| Workbook area | Product interpretation |
|---|---|
| **Customer** | Existing CRM customer master record. User selects from database only; no create/edit in this module. |
| **Project** | User can reuse an existing project under the selected customer or create a new one. Project name must be unique per customer. |
| **Comments** | Multiline free-text note field. |
| **Orange block** | Yes/No qualification questions that feed chance-to-win logic. |
| **Yellow block** | Pipeline stage tracking: Before, Approaching, Sent. Only one stage should be active at a time. |
| **Green block** | Amount breakdown by work type. Currently hardcoded as columns; should become repeatable amount-line records backed by an admin-managed type table. |
| **Final stage** | Due date plus perpetual / ongoing contract handling. |
| **Offer status** | Final outcome: Win, Lose, Suspended, Cancelled. Blank or Open means still active. |
| **Summary rows under the data** | Should become dashboard views and drill-down analytics, not stored as lead records. |

| Current-state finding | Implication |
|---|---|

| | |
|---|---|
| **Finding** | Summary rows and salesperson grouping rows are mixed into the same grid as real leads; the product must separate transactional data from reporting data. |
| **Finding** | Several hidden validation and summary cells already show formula breakage (#REF), which is a strong reason to move calculations into code and database queries. |
| **Finding** | The existing work-type list is useful as a seed list, but its fixed-column design is too rigid for productization. |
| **Finding** | The workbook already implies the need for live team comparison, customer/project comparison, and monthly/quarterly forecasting. |

Important design decision: stage must be modeled as a single-select field in the product, not as three separate spreadsheet-style yes/no cells. This removes the need for fragile validation formulas.

# 4. Roles and Personas

| Role | Primary responsibility |
|---|---|
| **Salesperson** | Creates and updates leads, answers chance questions, sets stage, enters amounts, and reviews personal dashboard metrics. |
| **Sales Manager** | Reviews team pipeline, compares salespeople, customers, and projects, and drills down into overdue or high-value opportunities. |
| **Admin** | Maintains reference data such as work types; optionally maintains calculation configuration if that scope is approved. |

# 5. Use Case Catalog

| ID | Use case | Primary actor | Business outcome |
|---|---|---|---|
| **UC-01** | Open the module from the host CRM | Salesperson / Manager | User enters the embedded module with a valid CRM session and token. |
| **UC-02** | Create a new lead | Salesperson | A new opportunity is captured against an existing customer. |
| **UC-03** | Reuse or create a project under a customer | Salesperson | Project is selected from suggestions or created if unique. |
| **UC-04** | Maintain qualification answers and stage | Salesperson | Chance-to-win inputs and stage stay current. |
| **UC-05** | Maintain amount lines | Salesperson | Work-type amounts are entered and totaled automatically. |
| **UC-06** | Set final stage data and close or suspend a lead | Salesperson / Manager | Due date, perpetual flag, and final outcome are updated. |
| **UC-07** | Review dashboard and drill into lead lists | Salesperson / Manager | Users consume KPIs, charts, and detail tables. |
| **UC-08** | Manage reference data | Admin | Admin maintains work types and, if approved, calculation configuration. |
| **UC-09** | Migrate legacy spreadsheet records | Admin / Project team | Existing lead rows are imported as normalized records. |

## 5.1 UC-01 - Open the module from the host CRM

| Item | Detail |
|---|---|
| **Primary actor** | Salesperson or Manager |
| **Goal** | Access the lead-management module without a separate login. |
| **Preconditions** | User is already authenticated in the CRM; CRM is configured to host the module. |
| **Main flow** | 1) User opens the module from a CRM menu or from a customer context page. 2) Host CRM loads the embedded client. 3) Host sends a JWT to the client by postMessage. 4) Client validates the sender origin and uses the token as a Bearer token for API calls. 5) If customer context was supplied by the CRM, the customer is pre-filled. |
| **Exceptions / rules** | If the token is missing, invalid, or expired, the module shows a recoverable message and asks the user to reopen from CRM or refresh the host context. The module does not show its own username/password screen. |

## 5.2 UC-02 - Create a new lead

| Item | Detail |
|---|---|
| **Primary actor** | Salesperson |
| **Goal** | Capture a new opportunity with the minimum required business context. |
| **Preconditions** | Customer exists in CRM master data; user has permission to create leads. |
| **Main flow** | 1) User clicks New Lead. 2) User selects customer (or customer is pre-filled from context). 3) User selects an existing project suggestion or enters a new project name. 4) User fills comments, chance questions, stage, amount lines, due date, and perpetual-contract flag as available. 5) System calculates totals and forecast values. 6) User saves the lead; system stamps owner, timestamps, and audit trail. |
| **Exceptions / rules** | Project name must be unique per customer. Formula-driven values cannot be edited. A lead may be saved with incomplete analytics inputs, but such a lead is clearly flagged as Incomplete and excluded from forecast metrics until the missing required fields are completed. |

## 5.3 UC-03 - Maintain qualification answers and stage

| Item | Detail |
|---|---|
| **Primary actor** | Salesperson |
| **Goal** | Keep the chance-to-win model up to date as the opportunity evolves. |
| **Preconditions** | Lead already exists and is open. |
| **Main flow** | 1) User opens an existing lead. 2) User answers or updates Yes/No qualification questions. 3) User selects exactly one stage: Before, Approaching, or Sent. 4) System recalculates chance-to-win and weighted forecast values immediately. 5) Updated values become visible in dashboards after save. |
| **Exceptions / rules** | Stage is mutually exclusive. If business approves configurable stage coefficients, the app reads them from configuration rather than code. If the lead is closed as Win/Lose/Cancelled, stage becomes read-only unless a manager reopens the lead. |

## 5.4 UC-04 - Maintain amount lines

| Item | Detail |
|---|---|
| **Primary actor** | Salesperson |
| **Goal** | Store the financial scope of the opportunity by work type. |
| **Preconditions** | Lead already exists; at least one active work type exists in reference data. |
| **Main flow** | 1) User adds one or more amount lines. 2) Each line requires work type and amount; note is optional. 3) Multiple lines may use the same work type if the user wants separate notes. 4) System totals all amounts by work type and also totals the whole lead. 5) Aggregated totals flow into dashboard analytics. |
| **Exceptions / rules** | Amounts must be positive numeric values. Deactivated work types remain visible on historical records but are unavailable for new entries. The total amount is always |

| | |
|---|---|
| | derived from lines and is never typed directly. |

## 5.5 UC-05 - Set final stage data and close or suspend a lead

| Item | Detail |
|---|---|
| **Primary actor** | Salesperson or Manager |
| **Goal** | Move an active lead into a final or paused outcome state. |
| **Preconditions** | Lead already exists. |
| **Main flow** | 1) User sets or updates due date. 2) User marks whether the lead is a perpetual contract. 3) User changes Offer Status from Open to Win, Lose, Suspended, or Cancelled when appropriate. 4) If the outcome is Win, the system may capture Actual Awarded Amount for forecast-vs-actual reporting. 5) Dashboards update by status and by period. |
| **Exceptions / rules** | Offer Status should default to Open. Suspended is reversible; Win/Lose/Cancelled are treated as closed states. If Actual Awarded Amount is adopted in MVP, it becomes required when Offer Status = Win. |

## 5.6 UC-06 - Review dashboards and drill into lists

| Item | Detail |
|---|---|
| **Primary actor** | Salesperson or Manager |
| **Goal** | Understand pipeline volume, forecast, risk, and performance without reading summary rows. |
| **Preconditions** | User has permission to view dashboards. |
| **Main flow** | 1) User lands on Dashboard. 2) User filters by owner, customer, project, work type, stage, due-date period, lead type, or offer status. 3) User reads KPI cards and chart widgets. 4) User clicks a chart slice/bar/row to open the filtered lead list or a specific lead. 5) User returns to dashboard without losing filters. |
| **Exceptions / rules** | Manager dashboards include team-comparison visuals. Salesperson dashboards default to the user's own pipeline. Filters should be persistent within the session. |

## 5.7 UC-07 - Manage reference data

| Item | Detail |
|---|---|
| **Primary actor** | Admin |
| **Goal** | Keep work-type options and, if approved, calculation configuration current without database scripts. |
| **Preconditions** | Admin permission is present. |
| **Main flow** | 1) Admin opens Reference Data. 2) Admin creates, edits, sorts, activates, or deactivates work types. 3) If configuration scope is included, admin also adjusts qualification weights and stage coefficients. 4) Changes apply to new or edited leads without corrupting history. |
| **Exceptions / rules** | Deactivation must not rewrite historical values. Configuration changes should be audited, because they affect forecast calculations. |

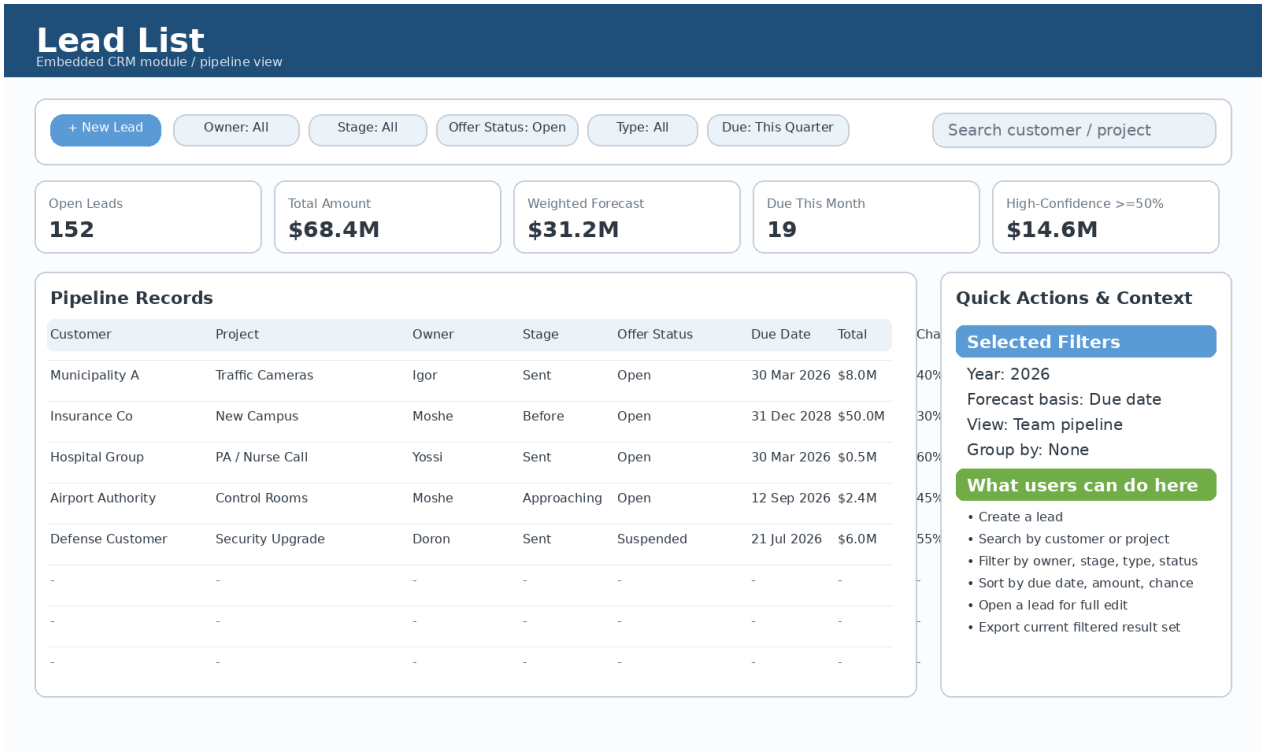
## 6. Information Architecture and Screen Map

| Screen | Primary users | Purpose |
|---|---|---|
| **Dashboard** | Salesperson / Manager | Default analytics view with filters, KPI cards, charts, and drill-down. |
| **Lead List** | Salesperson / Manager | Operational grid for search, sort, filter, and open/edit. |
| **Lead Create / Edit** | Salesperson / Manager | Structured form for all lead inputs plus live calculated summaries. |
| **Reference Data / Admin** | Admin | Work-type management and optional calculation configuration. |

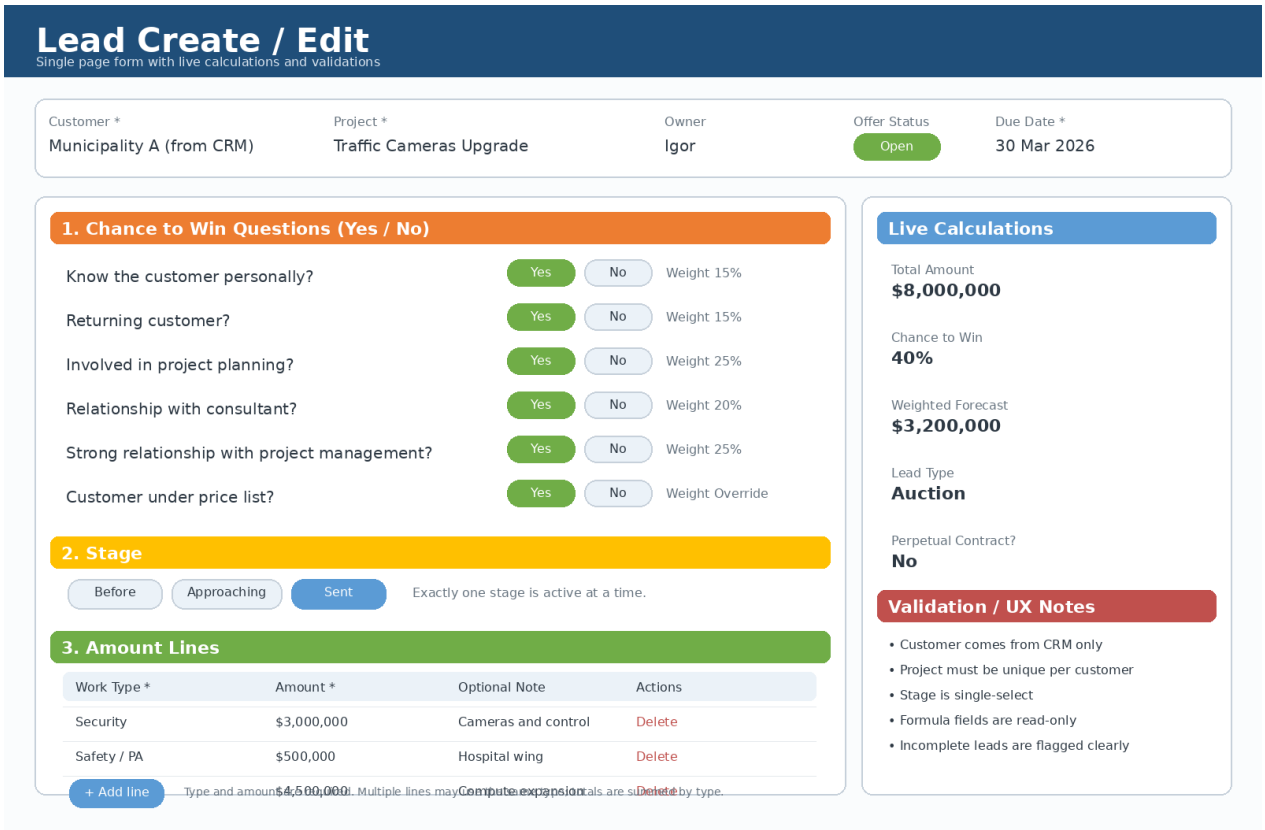
Recommended navigation flow: Dashboard -> Lead List -> Lead Create/Edit. Admin is a separate navigation item and should not be visible to standard sales users.

## 7. Low-Fidelity Wireframes

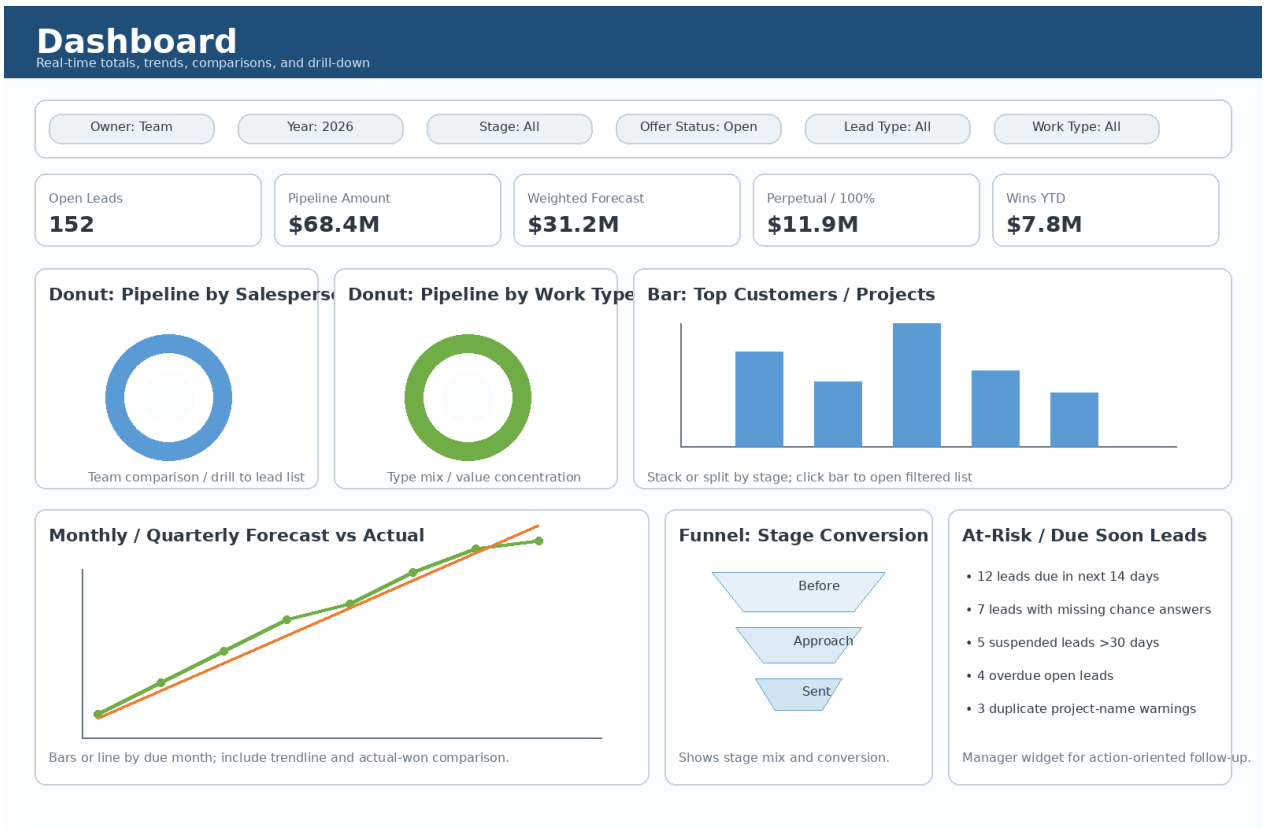
The wireframes below are intentionally low-fidelity. They define layout, interaction zones, and information hierarchy rather than final visual design.



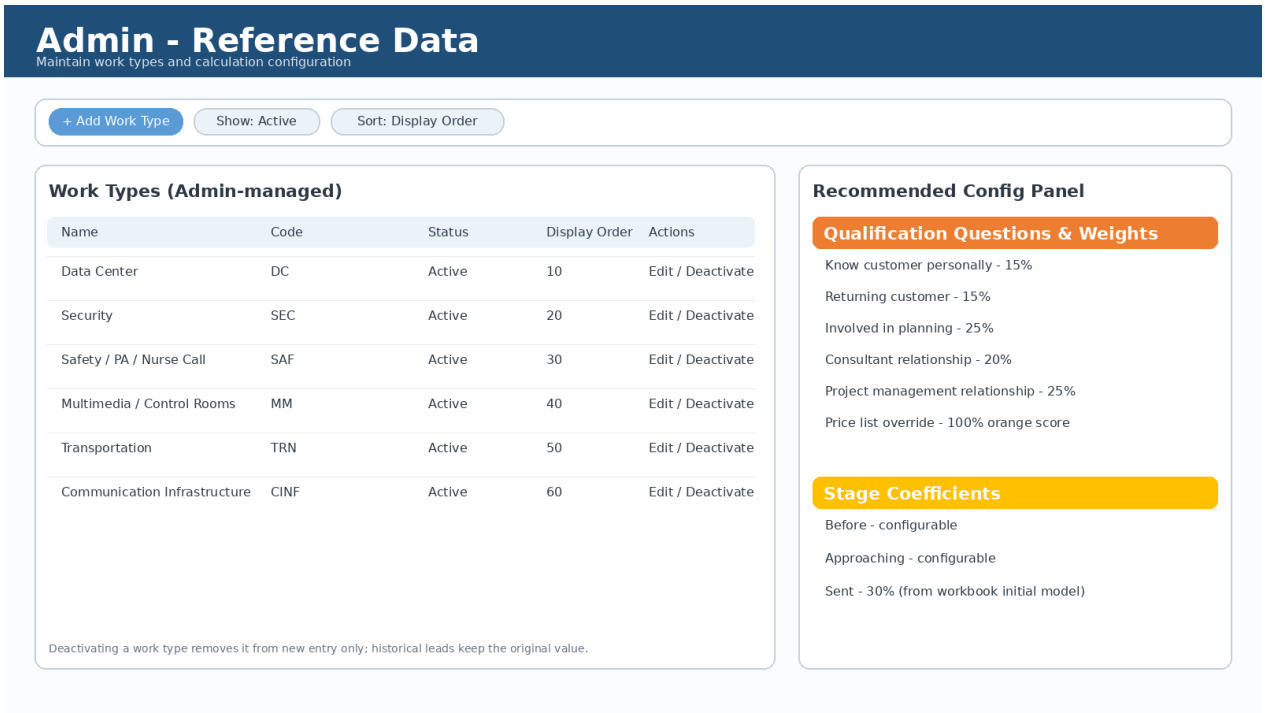

*Wireframe A - Lead List with filter strip, KPI summary, pipeline grid, and contextual action panel.*




*Wireframe B - Lead Create/Edit with guided sections and a live calculation sidebar.*




*Wireframe C - Dashboard with KPI cards, comparison charts, trendline, funnel, and action-oriented risk widget.*




*Wireframe D - Admin reference-data screen for work types and, optionally, calculation configuration.*

# 8. Functional Requirements

The requirements below are written so engineering can translate them into epics, stories, and acceptance criteria without having to infer spreadsheet behavior from formulas.

## 8.1 Lead List and Navigation

| ID | Requirement |
|---|---|
| **FR-LIST-01** | The module must provide Dashboard and Lead List as first-class views inside the CRM-hosted app. |
| **FR-LIST-02** | Lead List must support filtering by owner, customer, project, work type, lead type, stage, offer status, due date range, and amount range. |
| **FR-LIST-03** | Lead List must support search by customer name and project name. |
| **FR-LIST-04** | Lead List must support sorting by due date, last updated date, total amount, weighted forecast amount, and chance-to-win percentage. |
| **FR-LIST-05** | Clicking a row opens the lead in view/edit mode without exposing formula columns or spreadsheet-like free cell editing. |
| **FR-LIST-06** | The system should preserve current filters while the user drills from Dashboard to Lead List and back. |

## 8.2 Lead Record Management

| ID | Requirement |
|---|---|
| **FR-LEAD-01** | Customer is selected from CRM data only and cannot be created inside this module. |
| **FR-LEAD-02** | If the module is opened in customer context from the CRM, customer should be pre-filled and may be locked by configuration. |
| **FR-LEAD-03** | Project field must support autocomplete based on existing projects for the selected customer. |

| | |
|---|---|
| **FR-LEAD-04** | Users may create a new project name if no suitable suggestion exists; uniqueness must be enforced per customer. |
| **FR-LEAD-05** | Comments must support multiline text. |
| **FR-LEAD-06** | Owner should default from the authenticated CRM user and be editable only for authorized roles. |
| **FR-LEAD-07** | Offer Status values must be Open, Win, Lose, Suspended, and Cancelled. Open is the default state. |
| **FR-LEAD-08** | Due date and perpetual-contract flag belong to the lead record and must be editable in the form, not calculated. |
| **FR-LEAD-09** | The application must show created/updated timestamps and retain an audit history of changes. |
| **FR-LEAD-10** | Closed records should not be hard-deleted from the UI; use archive / close semantics instead. |

## 8.3 Qualification, Stage, and Amount Entry

| ID | Requirement |
|---|---|
| **FR-Q-01** | Qualification questions are presented as explicit Yes/No controls, not freeform numeric cells. |
| **FR-Q-02** | The initial qualification question set should reflect the workbook: knows customer personally, returning customer, involved in planning, consultant relationship, project-management relationship, and customer under price list. |
| **FR-STAGE-01** | Stage must be a single-select field with values Before, Approaching, and Sent. |
| **FR-STAGE-02** | The UI must prevent multiple active stages on one lead. |
| **FR-AMT-01** | Amounts are entered as repeatable child lines with mandatory work type and amount, and optional note. |
| **FR-AMT-02** | The work-type list must come from a database table managed by admin. |
| **FR-AMT-03** | The system must total all lines for the same work type and expose those totals to dashboards. |
| **FR-AMT-04** | The lead's total amount must be calculated from amount lines and never typed directly by the user. |

## 8.4 Outcomes, Forecast, and Dashboard

| ID | Requirement |
|---|---|
| **FR-DASH-01** | Dashboard must expose both raw pipeline amounts and calculated / weighted forecast amounts. |
| **FR-DASH-02** | Dashboard must support comparison by salesperson, customer, project, work type, stage, and offer status. |
| **FR-DASH-03** | Dashboard must support monthly and quarterly aggregation by due date. |
| **FR-DASH-04** | Charts must be drillable to filtered lead lists. |
| **FR-DASH-05** | Dashboard must include at least one trend chart that compares forecast and actual results over time. |
| **FR-DASH-06** | Dashboard must include a risk-oriented widget for due-soon, overdue, suspended, or incomplete leads. |

## 8.5 Admin and Configuration

| ID | Requirement |
|---|---|
| **FR-ADM-01** | Admin must be able to create, edit, sort, activate, and deactivate work types. |
| **FR-ADM-02** | Deactivating a work type must not change historical lead records. |
| **FR-ADM-03** | If approved for MVP, qualification weights and stage |

| | |
|---|---|
| | coefficients should be config-driven rather than hardcoded. |
| **FR-ADM-04** | Admin changes to reference data and configuration must be audited. |

# 9. Field Dictionary and Validation Rules

## 9.1 Core Lead Fields

| Field | Type | Source | Required | Validation / note |
|---|---|---|---|---|
| **Lead Owner** | User reference | JWT/CRM user or manager reassignment | Required | Defaults from CRM user; edit permissions by role. |
| **Customer** | Reference | CRM customer master | Required | Read-only lookup from CRM; no free typing. |
| **Project** | Text/reference | Existing project suggestion or new value | Required | Unique per customer after trim/case normalization. |
| **Comments** | Multiline text | User input | Optional | Recommended max length 2,000-4,000 chars. |
| **Stage** | Enum | User input | Required for forecast inclusion | Exactly one of Before / Approaching / Sent. |
| **Perpetual Contract?** | Boolean | User input | Required for forecast inclusion | Yes = perpetual/ongoing; No = auction/one-time. |
| **Due Date** | Date | User input | Required for forecast inclusion | Used for monthly/quarterly reporting. |
| **Offer Status** | Enum | User input | Required | Default Open; Win/Lose/Suspended/ Cancelled close or pause the lead. |
| **Actual Awarded Amount** | Currency | User input | Recommended when Win | Useful for forecast-vs-actual reporting. |

## 9.2 Qualification Questions (initial set)

| Question | Type | Initial weight | Notes |
|---|---|---|---|
| **Know the customer personally?** | Yes/No | 15% | Initial workbook weight. |
| **Returning customer?** | Yes/No | 15% | Initial workbook weight. |
| **Involved in project planning?** | Yes/No | 25% | Initial workbook weight. |
| **Relationship with consultant?** | Yes/No | 20% | Initial workbook weight. |
| **Strong relationship with project management?** | Yes/No | 25% | Initial workbook weight. |
| **Customer under price list?** | Yes/No | Override | Recommended to keep as a special business rule rather than a normal additive weight. |

## 9.3 Amount Line Fields

| Field | Type | Required | Rule |
|---|---|---|---|
| **Work Type** | Reference | Required | Value from active admin-managed work-type table. |
| **Amount** | Currency | Required | Positive number; zero and negative values rejected. |
| **Line Note** | Text | Optional | Free text that explains sub- |

| | | | |
|---|---|---|---|
| | | | scope for that line. |

### 9.4 Key Validation Rules

| ID | Validation |
|---|---|
| **VAL-01** | Customer must exist in CRM and be selected, not typed. |
| **VAL-02** | Project must be unique per customer, case-insensitive after trim. |
| **VAL-03** | Exactly one stage must be active. |
| **VAL-04** | At least one amount line is required before a lead enters forecast metrics. |
| **VAL-05** | Type and Amount are mandatory on each amount line; note is optional. |
| **VAL-06** | Calculated fields are read-only in the UI and server-authoritative in the API. |
| **VAL-07** | Leads missing required analytics fields must be flagged as Incomplete and excluded from forecast charts until completed. |
| **VAL-08** | If Actual Awarded Amount is included in scope, it should become mandatory when Offer Status = Win. |

## 10. Calculation Rules and Business Logic

The application should not replicate spreadsheet formulas literally. It should preserve the business logic in a more robust, configurable form. All calculations should run on the server; the client may preview values but cannot override them.

| ID | Metric | Definition | Notes |
|---|---|---|---|
| **CALC-01** | Total Amount | Sum of all amount-line amounts on the lead. | Server-derived; read-only. |
| **CALC-02** | Amount by Work Type | Sum of amount lines grouped by work type. | Used for dashboards and type breakdown. |
| **CALC-03** | Qualification Score | If Price List = Yes, treat orange-section score as 100%; otherwise sum Yes answers by configured weights. | Reflects initial workbook behavior. |
| **CALC-04** | Qualification Contribution | Qualification Score x orange-section coefficient. | Initial workbook coefficient is 30%. |
| **CALC-05** | Stage Contribution | Coefficient mapped from selected stage. | Workbook clearly uses a 30% stage coefficient for Sent; Before/Approaching coefficients should be confirmed or configured. |
| **CALC-06** | Calculated Chance to Win | Min(100%, Qualification Contribution + Stage Contribution). | Recommended for auction / one-time leads. |
| **CALC-07** | Forecast Amount | If Perpetual Contract = Yes, Forecast Amount = Total Amount; else Total Amount x Calculated Chance to Win. | Keeps ongoing/perpetual contracts at 100% value. |
| **CALC-08** | High-Confidence Forecast | Forecast Amount only when chance >= 50%, or perpetual contract = Yes. | Useful because the workbook already hints at a >50% management lens. |
| **CALC-09** | Won Amount | Actual Awarded Amount when Offer Status = Win; otherwise 0. | Supports forecast-vs-actual reporting. |

| Configuration | Initial recommendation |
|---|---|

| | |
|---|---|
| **Orange-section coefficient** | 30% |
| **Question weights** | 15%, 15%, 25%, 20%, 25%, plus Price List override |
| **Stage coefficients** | Recommended config-driven. Safe starting point: Before 0%, Approaching 15%, Sent 30%. |

Best-practice recommendation: make qualification weights and stage coefficients data-driven from day one. The workbook already demonstrates that these values are business settings, not a UI concern.

## 11. UX / UI Design Principles

| ID | Design principle |
|---|---|
| **UX-01** | Use a guided single-record form instead of spreadsheet-like row editing. |
| **UX-02** | Keep a live calculation sidebar visible while editing so the user immediately understands the impact of answers, stage, and amounts. |
| **UX-03** | Use explicit Yes/No controls for qualification answers and a segmented control for stage. |
| **UX-04** | Clearly separate editable fields from calculated fields using layout and styling, not by hidden columns. |
| **UX-05** | Show validation inline near the field and also in a top summary when save is blocked. |
| **UX-06** | Support keyboard-friendly data entry because the primary users are operational sales staff. |
| **UX-07** | Design desktop-first, but remain responsive down to tablet width. |
| **UX-08** | Support Hebrew / RTL localization, local date formatting, and local currency formatting if the CRM requires it. |

## 12. Dashboard and Analytics Requirements

The dashboard replaces the workbook's summary rows and should cover both operational pipeline management and management reporting.

### 12.1 KPI Cards

| KPI | Definition |
|---|---|
| **Open Leads** | Count of currently open leads after filters. |
| **Pipeline Amount** | Sum of total amounts on filtered open leads. |
| **Weighted Forecast** | Sum of forecast amounts on filtered open leads. |
| **Perpetual / 100% Value** | Amount committed through perpetual / ongoing contracts. |
| **High-Confidence Forecast** | Forecast amount with chance >= 50% or perpetual flag. |
| **Wins YTD / Period** | Actual won amount within the selected period. |

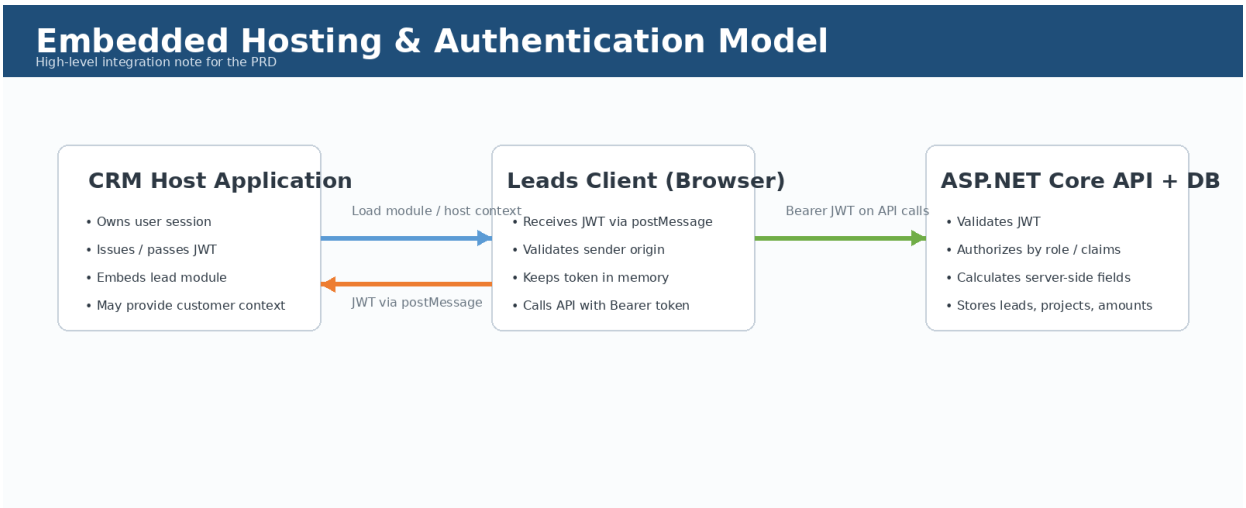
### 12.2 Minimum Chart Pack

| Chart type | Widget | Business purpose |
|---|---|---|
| **Donut / Pie** | Pipeline by salesperson | Team comparison and ownership visibility. |
| **Donut / Pie** | Pipeline by work type | Understand concentration of business by solution area. |
| **Bar** | Top customers by total or weighted amount | Compare customer value and drill to filtered records. |
| **Bar** | Top projects by total or weighted amount | Compare project opportunity scale. |
| **Column + Trendline** | Monthly / quarterly forecast by due date | Show forecast evolution over time. |
| **Line / combo** | Forecast vs actual won amount | Measure forecast quality against final |

| | | |
|---|---|---|
| | | result. |
| **Funnel** | Before -> Approaching -> Sent -> Won/Lost | Show stage distribution and conversion. |
| **Risk table** | Due soon / overdue / suspended / incomplete leads | Support day-to-day action by sales and management. |

## 13. Integration Architecture and Security Notes

The module is intentionally designed as an embedded application hosted by the existing CRM rather than as a standalone system.




*High-level hosting and authentication model captured for the PRD.*

| ID | Requirement / note |
|---|---|
| **INT-01** | The Leads Management module is a single ASP.NET Core web application backed by the database. |
| **INT-02** | The host CRM provides the authenticated context; this module does not implement a standalone login page. |
| **INT-03** | The client receives the JWT from the host application by postMessage. |
| **INT-04** | The client uses the received JWT as a Bearer token on requests to the ASP.NET Core backend. |
| **INT-05** | The client must validate allowed host origins before accepting postMessage payloads. |
| **INT-06** | The backend must validate token signature, expiry, audience, and authorization claims. |
| **INT-07** | If host CRM passes a customer context, the module should accept and use that context to pre-fill the form. |
| **INT-08** | For security hygiene, the preferred client behavior is to keep the token in memory rather than browser persistence. |

## 14. Conceptual Data Model

| Entity | Purpose | Key fields |
|---|---|---|
| **Customer** | External CRM master entity | CustomerId, Name, CRM metadata. Read-only from this module. |
| **Project** | Child entity under customer | ProjectId, CustomerId, Name, IsActive. Unique(CustomerId, ProjectName). |
| **Lead** | Core opportunity record | LeadId, ProjectId, OwnerUserId, Comments, Stage, DueDate, IsPerpetual, OfferStatus, ActualAwardedAmount, CreatedAt, UpdatedAt. |

| | | |
|---|---|---|
| **LeadAmountLine** | Amount breakdown child entity | LeadAmountLineId, LeadId, WorkTypeId, Amount, Note. |
| **WorkType** | Reference data | WorkTypeId, Name, Code, SortOrder, IsActive. |
| **QualificationQuestion** | Configuration entity (recommended) | QuestionId, Code, Label, Weight, SortOrder, IsActive, IsOverrideRule. |
| **QualificationAnswer** | Lead-specific answers | LeadId, QuestionId, AnswerYesNo. |
| **Audit / History** | Change tracking | EventId, Entity, EntityId, ChangedBy, ChangedAt, OldValue, NewValue. |

## 15. Migration from the Existing Workbook

| ID | Migration rule |
|---|---|
| **MIG-01** | Import only true input rows as lead records. Salesperson header rows, grouped summary rows, and formula-only rows are excluded. |
| **MIG-02** | Treat calculated spreadsheet columns as non-authoritative. Recalculate values inside the new application after import. |
| **MIG-03** | Map fixed work-type columns in the workbook into normalized amount-line records. |
| **MIG-04** | Preserve source workbook metadata such as sheet name and source row number for auditability during migration. |
| **MIG-05** | Confirm the business meaning of the two workbook sheets before migration. They may represent categories, views, or legacy reporting partitions. |

Migration caution: because the workbook already contains broken hidden formulas in some validation/summary areas, the import should trust raw user-entered inputs rather than trying to preserve workbook-calculated values.

## 16. Recommended Release Slicing

| Release | Recommended scope |
|---|---|
| **MVP** | Embedded auth, lead CRUD, project uniqueness, qualification questions, stage, amount lines, due date, perpetual flag, offer status, server-side calculations, dashboard, work-type admin, audit trail. |
| **Phase 2** | Annual targets / quotas, automated reminders, exports, richer configuration admin, alerts on stale leads, deeper forecast-vs-actual analysis. |

## 17. Risks, Open Questions, and Recommended Decisions

| ID | Open question | Recommended direction |
|---|---|---|
| **Q-01** | What do the current workbook sheets represent in the new product model? | Confirm whether they become a lead category/tag, a default filter, or should disappear entirely. |
| **Q-02** | Should Before and Approaching stages contribute probability, or only Sent? | Use config-driven stage coefficients; default recommendation is 0% / 15% / 30%. |
| **Q-03** | Should Actual Awarded Amount be mandatory on Win? | Recommended yes if forecast-vs-actual reporting matters in MVP. |
| **Q-04** | Should visibility be own-leads only or team-wide by role? | Recommended: own leads for salespeople, team/global visibility for managers. |

| | | |
|---|---|---|
| **Q-05** | What currency and number formatting rules should be used? | Adopt CRM locale and make currency formatting consistent across list, form, and dashboard. |
| **Q-06** | Do you want annual targets / quotas in scope? | Recommended Phase 2 unless targets already exist in CRM and are easy to consume. |

## 18. MVP Acceptance Criteria

| ID | Acceptance criterion |
|---|---|
| **AC-01** | User can open the module from CRM without separate login and make authenticated API calls via Bearer token. |
| **AC-02** | Customer is selected from CRM data only; free-text customer entry is impossible. |
| **AC-03** | Project uniqueness is enforced per customer. |
| **AC-04** | Exactly one stage can be active on a lead. |
| **AC-05** | User can add multiple amount lines; system totals them by type and overall. |
| **AC-06** | Total amount and forecast amount are always calculated and read-only. |
| **AC-07** | Perpetual-contract leads can be represented and shown as 100% forecast value. |
| **AC-08** | Offer Status supports Open, Win, Lose, Suspended, and Cancelled. |
| **AC-09** | Dashboard shows KPI cards, comparison charts, trendline, and risk widget with drill-down. |
| **AC-10** | Admin can maintain work types without database scripts. |
| **AC-11** | Audit trail captures create/update/config changes. |

## 19. Final Recommendation

Build the module as a structured operational product, not as a spreadsheet clone. Keep the business language familiar to users, but model the data cleanly: one lead record, one stage, repeatable amount lines, read-only calculated metrics, and dashboards instead of summary rows.

If you want this to succeed with sales users, the most important UX decisions are: customer from CRM, project autocomplete plus unique create, a single-page form with a live summary sidebar, and dashboards that let users drill back into actionable lists.